भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

Indian Institute of Information Technology Guwahati

CS236 - Artificial Intelligence Lab
Assignment - 02

1. Develop a Python program to interact with a text file named "example.txt". The program should perform the following operations:

   i. Create a file named "example.txt" and populate it with some sample text. Ensure the file is saved in the same directory as your Python script.

   ii. Read the entire content of "example.txt" using Python and display it.

2. Write a Python program that reads a user-specified number of characters from a text file. Create a text file, and then your program should:

   i. Prompt the user to enter the number of characters to read.

   ii. Read and print the specified number of characters from the file.

3. Extend the program from Question 2 to append an additional line of text to the existing file. After appending, the program should display the entire updated content of the file.

4. Create a Python function named `count_lines_not_starting_with_T` that counts the number of lines in a text file named "text.txt" that do not begin with the letter "T". For example, assume "text.txt" contains the following lines:

```
A girl is playing there badminton.
The scenery is beautiful.
The birds are flying in the sky.
The sky is cloudy.
Alphabets consists of vowels and consonants.
```

5. Develop a Python program that reads the content of one text file and writes it to another. The program should:

   i. Read the content from an existing text file.

   ii. Create a new text file.

   iii. Write the content read from the first file into the newly created file.

6. You are given a NumPy array (matrix) of dimensions $N \times M$ with space-separated integer elements. Your task is to calculate and display the transpose and the flattened version of this matrix.

   i. Input the matrix from the user or define it in your code.

ii. Calculate and print the transpose of the matrix.
iii. Calculate and print the flattened version of the matrix.

7. Given a NumPy array of integers: `[10,16,71,9], [71,91,31,51]`, write a program to determine and print the following:
   i. The dimension of the array.
   ii. The shape of the array.
   iii. The size (total number of elements) of the array.

8. Create two random NumPy arrays of any dimension and perform the following operations:
   i. Concatenate the two arrays into a single array.
   ii. Sort both arrays individually.
   iii. Add the two arrays element-wise.
   iv. Subtract the second array from the first array element-wise.
   v. Multiply the two arrays element-wise.
   vi. Divide the first array by the second array element-wise (handle potential division by zero).

9. Generate a random NumPy matrix of size 8×7, where columns represent features and rows represent patterns. Find and display the maximum and minimum values within each feature (column).

10. Write a NumPy program to accomplish the following:
    i. Create a 4×5 matrix with values ranging from 1 to 20.
    ii. Create three arrays: one with 10 zeros, one with 10 ones, and one with 10 fives.
    iii. Create an array containing all even integers from 10 to 50.
    iv. Generate a single random number between 0 and 1.
    v. Save the 4×5 matrix created in part (i) to a text file named "matrix.txt". Then, load the matrix back from this file.